Insulation Paper Life Enhancement with the Natural Ester Fluid Biotransol – A Spectroscopic Investigation

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SYNOPSIS

Natural Esters have gained popularity in the recent years as Dielectric Fluids. This is due to their obvious advantages over mineral oils – high fire point and ready biodegradability. Besides these advantages, there is yet another performance related benefit in a transformer. This has to do with the enhancement of insulation paper life using this new fluid. This unexpected result has prompted several research groups to investigate the mechanism for this enhancement of paper life which will result in transformer asset life enhancement. In this paper, we present spectroscopic studies aimed at understanding the mechanism using Biotransol, a natural ester fluid developed at Savita. We have reported a mechanism using FTIR and XPS (X-Ray Photoelectron Spectroscopy) as supporting evidences.

KEYWORDS

Natural ester fluid (Biotransol-HF), enhancement of insulation life, thermallyupgraded kraft paper, transesterification, degree of polymerization, FTIR, XPS.

1.0 INTRODUCTION

Natural Esters are increasingly becoming a preferred fluid in transformers by virtue of their excellent fire-safety vis-à-vis mineral oils [1]. The fluid is also preferred at environmentally sensitive areas due to its ready biodegradability and non-toxicity to marine life [1, 2, 3]. Besides these benefits, there is yet another important aspect which is related to performance. This has to do with enhancement of life of paper insulation. This is a very significant benefit as the life of a transformer asset is largely determined by the life of paper insulation [4].

The question is what is the mechanism by which a natural ester slows down the ageing of paper compared to mineral oil? To answer this question, several laboratories have done scientific investigations of the two systems and have come up with a few speculations [2, 3, 4, 5, 6]. It is in this background that we, at Savita, have undertaken this work. Savita Oil Technologies has developed its own Natural ester, Biotransol HF (hereafter referred to as Biotransol in the text), and in this paper we report our investigations to explore and find mechanistic pathways contributing to slowing down of ageing of paper in presence of our natural ester.

2.0 EXPERIMENTAL DETAILS

2.1 Raw Materials

The insulating liquids used in this experiment were natural ester (Biotransol-HF) and mineral oil (Transol). The type of transformer insulation paper used for these studies was thermally upgraded Kraft paper (TUK) (Make - Weidman 22 HCC Insuldur). The dimensions of paper are thickness-0.09 mm and width-25.35 mm. The initial moisture content of thermally upgraded kraft paper was maintained as per IEEE C57.100–2011(about 0.25% to 0.5% by weight) [7].

2.2 Experimental Set-Up (Sealed Tube Reactor)

The Stainless Steel-316 sealed tube reactors are made as per dimensions provided in IEEE

C57.100 –2011 [7]. The reactor is provided with two valves – one for purging of nitrogen with dip tube and the other for vacuum and nitrogen venting. Pressure gauge is provided to monitor the pressure of reactor. The reactor is kept inside an electrically heated hot air oven for the duration of experiment. The fig.1 shows the reactor used. On completion of experiment, kraft paper was degreased by AR grade n-Hexane 4 to 6 times before carrying out spectroscopic work and measurement of degree of polymerization.

Fig.1 Sealed tube reactor as per IEEE C57.100-2011

2.3 Ftir Spectroscopy:

An ALPHA-ATR FTIR Spectrometer (Make - Bruker) was used to get infrared spectrum of functional groups of paper insulation. The attenuated total reflection (ATR – Diamond) technique was used to get the infrared spectra [8]. Paper samples, placed in the path of an infrared beam, will absorb and transmit light and the transmitted light signal is detected by the detector (DLATGS detector). This measurement needs only a small sample of insulating paper. In order to enhance the signal strength, the paper was folded 3 to 5 times before FTIR analysis. The FTIR wavelength range of 4000–500cm-1 was scanned for the reported spectra (in the transmission mode) [8].

2.4 XPS (X-Ray Photoelectron Spectroscopy)

A K-Alpha XPS spectrometer (Make – Thermo Scientific) was used for surface analysis of the new and aged papers. In the XPS technique, an X-ray photon is used to knock off an electron from an inner orbital, say C1s in the case of carbon and the binding energy of the electron emitted is measured in a detector. Chemical shifts in the binding energy of an element are associated with a particular chemical structure. For instance, Carbon 1s electron binding energy is sensitive to its environment and changes depending on its presence as carbonyl or ether etc [9].

2.5 Degree of Polymerization Measurement

The numbers of monomer units in cellulose are measured as the degree of polymerization (DP) of the cellulose polymer. Standard test method for determination of average viscometric degree of polymerization (abbreviated DPv) of new or aged electrical papers is ASTM D 4243–99/IEC 60450. The determination is made by measuring the intrinsic viscosity of a solution of the paper in an appropriate solvent (Cuen solvent). The degree of polymerization of a particular cellulose molecule is the number of anhydro-β-glucose monomers, C6H10O5. Within a sample of paper, not all the cellulose molecules have the same degree of polymerization so that its mean value is measured by viscometric methods [10].

3.0 RESULTS & DISCUSSIONS

3.1 Comparison of Visual Appearance of Paper Samples

Accelerated ageing studies of paper in fluids were conducted at a very high temperature of 1800C for the purpose of comparing appearance and rate of degradation of paper in mineral oil and our natural ester, Biotransol. Such a high temperature was chosen to hasten the process of degradation so that one can obtain comparative results in reasonable time. On the other hand, for spectroscopic studies, a slightly lower temperature of 1650C was chosen to ensure that surface species due to interaction between paper and oil are well preserved in the spectrometer. The fig.2 shows a comparison of visual appearance of mineral oil-aged versus

Biotransol-aged paper sample at different time intervals.

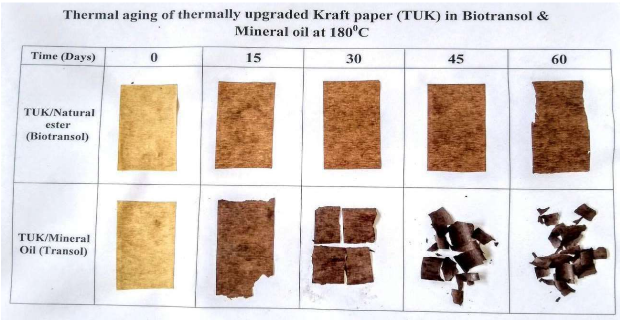


Fig. 2 Visual appearance aged kraft paper in Biotransol and Mineral oil.

At 180^{0}C, mineral oil-aged paper begins to show degradation even in 15 days. By the 15th day of aging, the paper begins to crumble in mineral oil whereas the corresponding paper aged in Biotransol appears robust even after 45 days of aging.

3.2 Comparison of Degree of Polymerization:

The differences in appearance are further corroborated by measurement of degree of polymerization (DP). The table1 shows the variation of DP values with time while the fig.3 shows the graphical plot of the values of degree of polymerization (DP) for the above samples.

Natural ester (Biotransol)/TUK	DP Value
Times (days)	180^{0}C
0	1100
15	378
30	312
45	246
60	217

Mineral oil(Transol)/TUK	DP Value
Times (days)	180^{0}C
0	1100
15	182
30	154
45	Too low
60	-

Table 1 Values of degree of polymerization of thermally upgraded kraft paper aged in Biotransol and mineral oil with respect to time and temperature

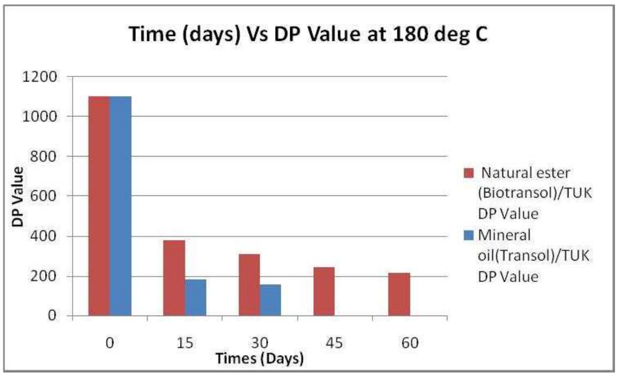


Fig. 3 Graphically representation of TUK degradation with respect to time in natural ester and mineral oil at 1800C

The values clearly show that Biotransol-aged paper has life even after 60 days with a DP of 217 while the paper aged in mineral oil degraded in 15 days itself with a DP value of 182. (A DP value of 200 is generally considered as the cut off point for the paper insulation life).

3.3 Structural Changes of Transformer Insulation Paper by Ftir & XPS Spectroscopy

3.3.1 Ftir

FTIR spectra were taken for fresh untreated TUK paper and TUK paper samples aged in Biotransol at 165^{0}C. This was carried out to see if any chemical species are formed on the paper surface during the ageing process. Interestingly, the fig. 4 shows that after 150 days (3600 hrs) of aging in Biotransol, the paper exhibits an absorption peak at 1743 cm^{-1} (Black color spectrum). This peak can be seen to be absent in the fig. 4 for fresh TUK paper (Red color spectrum). This peak was also absent in mineral oil aged paper (the spectrum for mineral oil aged paper is not shown in the fig.4).

What is the interpretation of this peak at 1743.13 cm^{-1}? Evidently, this is due to the formation of ester group on the paper surface. This could be caused by transesterification of cellulosic hydroxyl groups with the natural ester Biotransol.

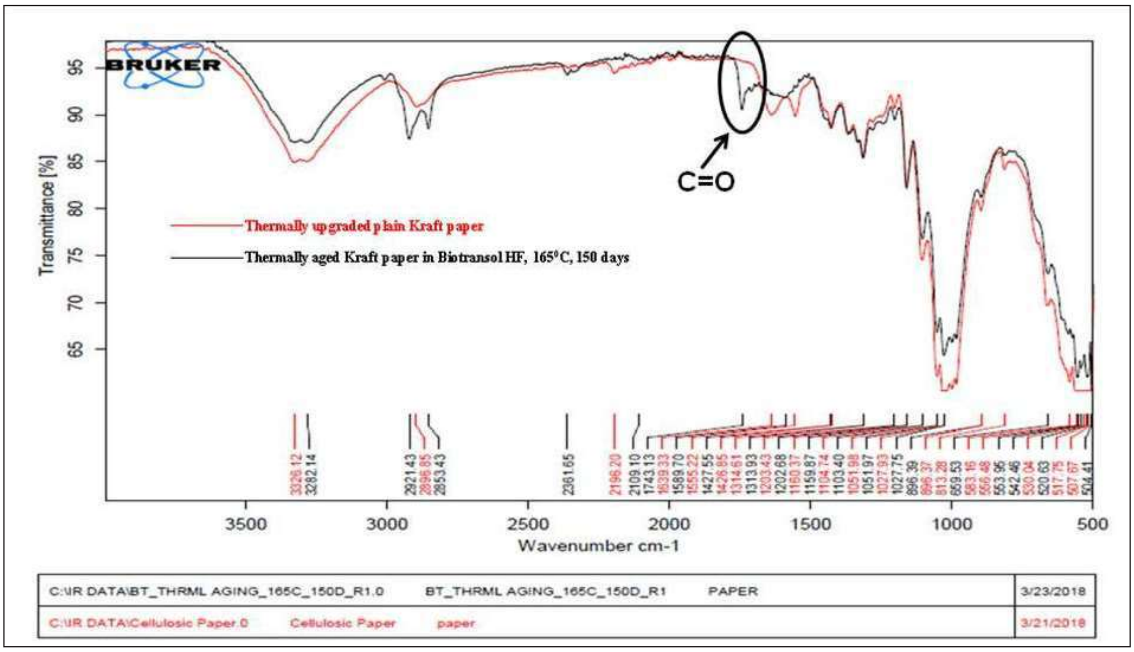


Fig. 4 Comparison of FT IR spectra of plain thermally upgraded kraft paper with aged paper

3.3.2 X-Ray Photoelectron Spectroscopy (Xps)

Further corroboration of FTIR was carried out using XPS technique. XPS is a powerful surface spectroscopy technique and can detect surface ester groups from the typical binding energy values of electrons in different orbitals. The binding energy of C1s is sensitive to its chemical environment. This means that a chemical shift observed in the binding energy of C1s electron can be correlated to the presence of either a carbonyl or other functionalities. The table 2, taken from literature [9], shows the correlation between binding energies of C1s electron and the chemical states.

Chemical state	Binding energy (C1s/eV)
C-C	284.8
C-O-C	~286
O-C=O	~288.5

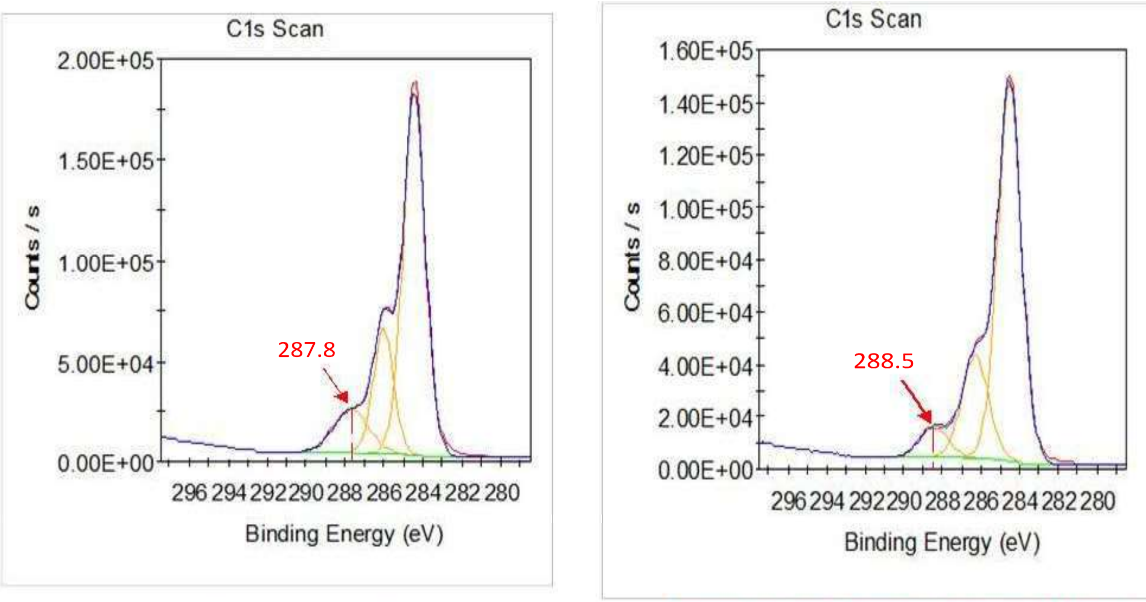
Table No.2 Binding energies of common chemical states

XPS spectra of the paper treated by Biotransol at 165^{0}C were compared with an untreated paper in the fig.5

A peak at 288.5 eV in the Biotransol-aged paper confirms the presence of carbonyl due to an ester group. Therefore XPS confirms our hypothesis about transesterification reaction taking place between the natural ester and the cellulosic paper. This interpretation is based on binding energy values reported in the literature for common chemical states of carbon (refer table no.2).

4.0 Our Proposed Mechanism for insulation life enhancement

Based on FTIR and XPS results, our proposal for mechanism of paper life enhancement is as follows: The natural ester, Biotransol, is extracting moisture from paper insulation and getting hydrolyzed to diglyceride and fatty acid. These free fatty acids, in turn, may be reacting with cellulosic hydroxyl groups to form an ester group on paper. We speculate that the paper degradation is slowing down due to the following three factors: Firstly, the esterified cellulose could increase the tensile strength of the paper by a simple increase in molecular weight of paper. Secondly since the moisture in the paper is extracted by the ester oil, the paper remains relatively dry in comparison to a mineral oil- aged paper. Thirdly, due to transesterification reaction, cellulose would become more hydrophobic and keep moisture away. This mechanism is shown schematically in fig. 6



Plain Thermally upgraded Kraft paper

Biotransol-aged Thermally upgraded Kraft paper for 150 days at 165°C

Fig No.5 Comparative values of Binding Energy of TUK Paper by XPS

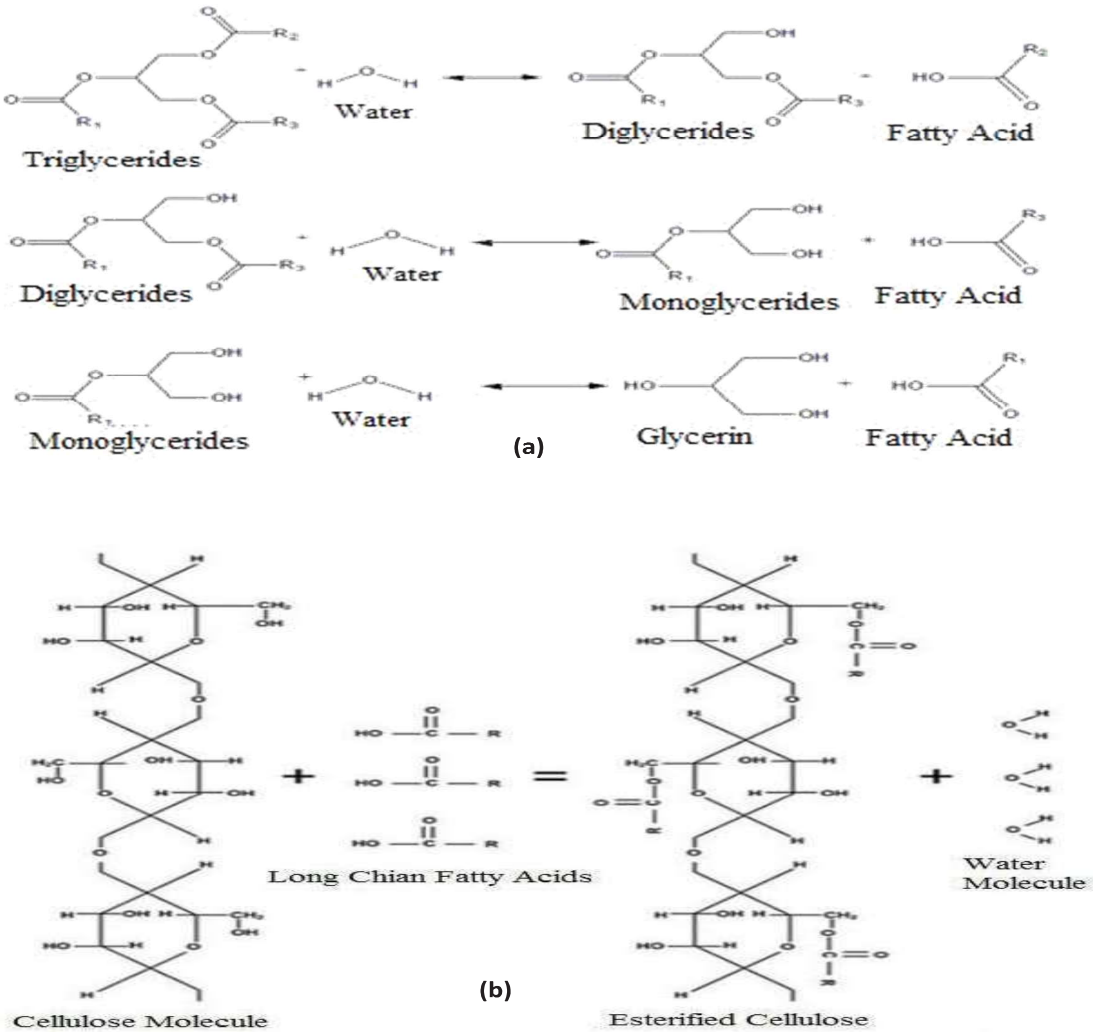


Fig No.6 (a) Hydrolysis of Natural Ester (b) Esterification of cellulose paper

5.0 SUMMARY & CONCLUSIONS:

Spectroscopic evidence was presented in this article on our natural ester fluid Biotransol in order to investigate how the ester fluid enhances the life of paper insulation in comparison to conventional mineral oils. The study involved sealed tube experiments conducted over a period of a few months with mineral oil and Biotransol. Besides visual appearance, FTIR and XPS of spectroscopic techniques were also used to understand the mechanistic pathways for slowing down of paper ageing phenomenon. Based on the spectroscopic evidence, a mechanism is proposed for the interaction between cellulosic hydroxyls and the natural ester. Transesterification reactions are proposed for strengthening the paper and slowing down its ageing in ester fluid.

6.0 REFERENCES

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